QNED vs QLED vs OLED TVs

What's the difference?

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If you are looking to upgrade your TV, terms like QNED, QLED, and OLED may sound similar but are quite different. While there are many details to address here, simplifying things for those not familiar with display jargon, we would start with the broader classification as self-emissive and backlit.

Self-emissive displays like OLED, QD-OLED, and micro-LED have pixels that emit their light, allowing for pixel-level lighting control, higher contrast, and brightness. These displays are relatively more expensive, and the general consensus is that they offer a better TV viewing experience.

On the other hand, when discussing QLED, QNED, Triluminous, and other such technologies, we are essentially referring to LCD TVs with a backlight. Manufacturers employ various technologies to improve the performance of these backlit displays and classify them with different names.

Why do LED LCD TVs need to enhance the quality of white light?

As you must know, colourimetry follows a trichromatic approach where white light is considered a combination of Red, Green and Blue light. LCD-based TVs eventually pass the white light from the backlight through a colour filter film to get Red, Blue and Green colours for different subpixels. So, to get the red colour, the filter will essentially block Red and Blue components in the white light and only let the red light to pass through (with some loss in quantum efficiency).

But what if your white light has more of the blue component and very less of Red and Green? In such cases, the TV struggles to reproduce the complete spectrum of Red and Green, especially the fully saturated shades.

To work around this problem, QLED TVs use Blue LEDs in the backlight and pass it through a Quantum Dot Enhancement Film (QDEF). This film has a matrix of tiny Quantum Dots that serve as extremely efficient colour converters. The blue light from the backlight falling on these Red and Green quantum dots excites them and they then release red and green light of the same quantum efficiency. So, the Blue light is essentially converted to red or green without any loss.

QDEF thus converts a balanced portion of Blue light to Red and Green, thus achieving a 'purer' or well-balanced white light with proper spectral power distribution, fit to reproduce wide colour gamuts.

Implementing Quantum Dot hardware in an LCD TV is quite simple. In practice, if you teardown your QLED TV, Quantum Dot Enhancement Film will be just a thin, yellowish butter paper-like film sandwiched between other backlight films (like diffuser film, brightness enhancement film, etc).

Display manufacturers may use other techniques to achieve a more balanced white light like a phosphor layer next to LEDs. And that is one of the techniques Sony integrates into its Triluminous displays.

QNED Vs QLED

Now that we have cleared the basics, we are in a better position to differentiate between QNED and QLED TVs.

QLED is a Samsung trademark for its TVs that use Quantum Dots Enhancement Film. The term isn't exclusive to Samsung since many other brands also use the QLED moniker for marketing their panels using the same technology. Legacy brands like LG and Sony, however, steer clear of "QLED" and have their proprietary technologies for enhancing backlight.

LG QNED stands for Quantum-Dot NanoCell LED and it combines Quantum Dots Enhancement Film with LG's Nanocell technology.

Apart from Quantum Dot Enhancement Film that all QLED TVs use, LG QNED also includes Nanocell technology that integrates nanoparticles in LCD substrate to absorb impure wavelengths from the backlight, thus once again resulting in purer R, G, and B components and purer colours.

Samsung denotes QLED TVs with mini LED backlights as Neo QLED TVs. LG QNED TVs may have both LED or Mini LED backlight depending on the